

Exploration in RL: Multi-armed Bandit (Continue)

**Kianté Brantley
CS 1840 Fall 2026**

Announcements

HW1 is out, and it's due today (**09/28 @ 12 midnight**)!

HW2 released last Friday!

Slight relaxation of HW late hour policy (**please check course website**)

My office hours are **canceled** today.

Check the course website for Office Hours:

<https://harvard-cs-1840.github.io/2026/>

Recap: MAB

Interactive learning process:

For $t = 0 \rightarrow T - 1$

(# based on historical information)

1. Learner pulls arm $a_t \in \{1, \dots, K\}$
2. Learner observes an i.i.d reward $r_t \sim \nu_{a_t}$ of arm a_t

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Learning metric:

$$\text{Regret}_T = T\mu^\star - \sum_{t=0}^{T-1} \mu_{a_t}$$

Handwritten notes:
- A red underline is under Regret_T .
- A red arrow points up to $\mu^\star$.
- To the right, red text says "lim $\rightarrow 0$ ".
- Further right, red text says " $\frac{\text{Regret}}{T} \rightarrow 0$ ".

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The Explore and Commit Algorithm:

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The Explore and Commit Algorithm:

For $k = 1 \rightarrow K$: (# Exploration phase)

Pull arm- k N times, observe $\{r_i\}_{i=1}^N \sim \nu_k$


Calculate arm k 's empirical mean: $\hat{\mu}_k = \sum_{i=1}^N r_i / N$


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For $t = \overline{NK} \rightarrow T - 1$: (# Exploitation phase)

Pull the best empirical arm, i.e., $a_t = \arg \max_{i \in [K]} \hat{\mu}_i$

Statistical Tools:

1. Hoeffding inequality (optional, no need to remember or understand it)

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$\{r_i\}_{i=1}^N \sim \nu$, w/ probability at least $1 - \delta$ we have:

$$\left| \sum_{i=1}^N \underset{\Delta}{r_i} / N - \underset{\Delta}{\mu} \right| \leq O \left(\sqrt{\frac{\ln(1/\delta)}{N}} \right)$$

$$|\hat{\mu} - \mu| \leq O \left(\sqrt{\frac{\ln(1/\delta)}{N}} \right)$$

$$|\hat{\mu} - \mu| \leq O \left(\sqrt{\frac{6}{N}} \right)$$

$$\Pr(\mu \notin CI) \leq \delta$$

$$\Pr(\mu \in CI) \geq 1 - \delta$$

$$\delta = .001$$

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i.e., this gives us a confidence interval:

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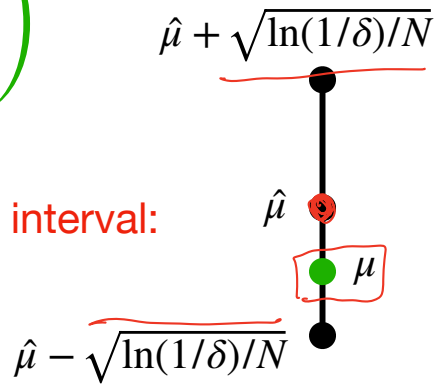
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After the Exploration phase, with probability at least $1-\delta$, **for all**

Δ **arm** $i \in [K]$, we have:

$$\left| \sum_{i=1}^N \underbrace{r_{i,n}}_{\mu_i} / N - \mu_i \right| \leq O \left(\sqrt{\frac{\ln(1/(\delta/K))}{N}} \right)$$

$$\delta = \frac{\delta}{K}$$

In summary, we have valid confidence intervals:

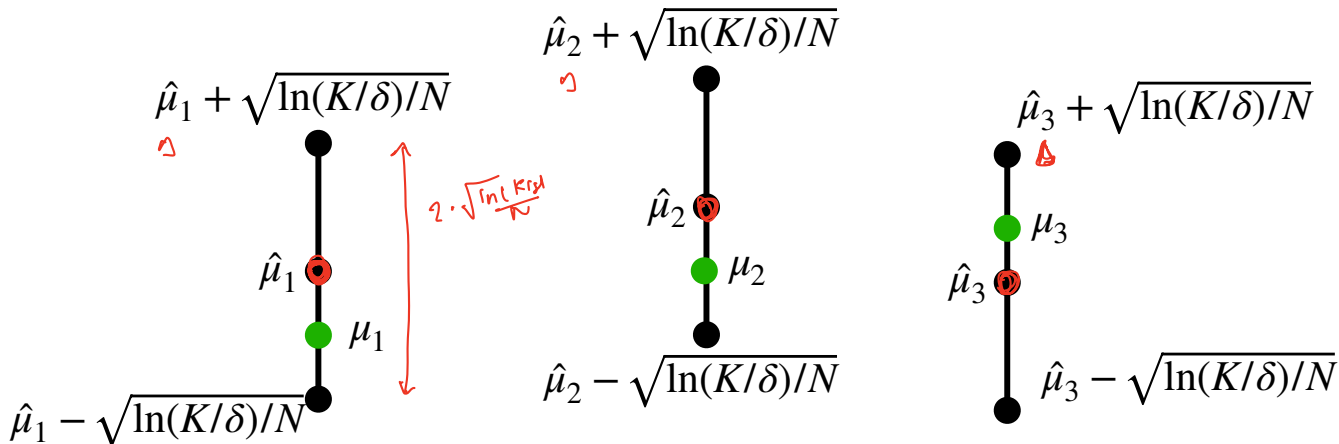
After the Exploration phase, with probability at least $1-\delta$, **for all**
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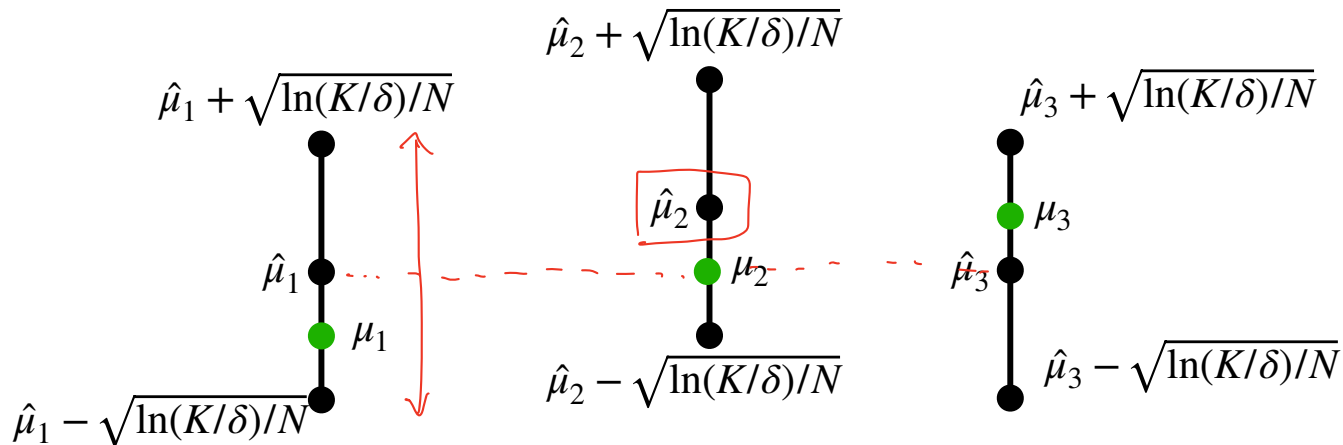
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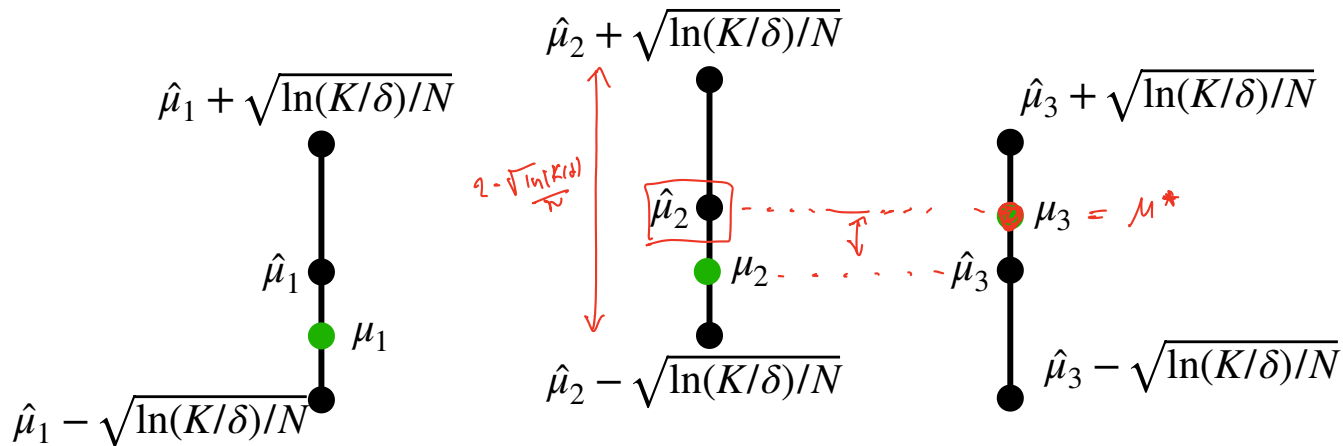
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In the rest, we will condition on the event that the confidence intervals are valid...



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Recall the Alg in this case will pick $a_t = 2$, for all $t \geq NK$,
(but it will suffer regret $(T - NK)(\mu_3 - \mu_2)$)

Calculate the final regret:

Denote **empirical best arm** $\hat{k} = \arg \max_{i \in [K]} \hat{\mu}_i$, and **THE best arm** $k^\star = \arg \max_{i \in [K]} \mu_i$

$\triangle$ *over estimator* $\triangle$ *Ground Truth*

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Handwritten notes:

- $\mu_{k^\star}$: Ground truth
- $\underline{NK}$: A round we commit to next empirical arm
- $\mu_{\hat{k}}$: commit to

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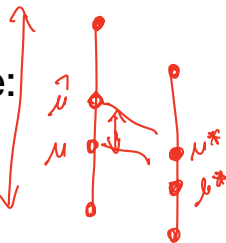
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Let's now bound $\text{Regret}_{\text{exploit}}$

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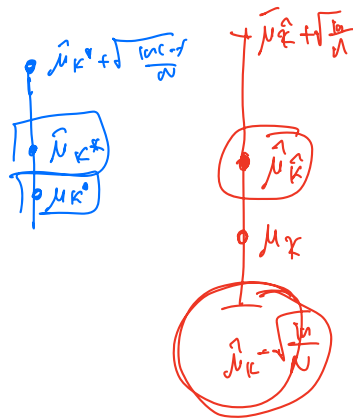
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What's the regret in the exploitation phase:

$$\text{Regret}_{\text{exploit}} \leq (T - NK)(\mu_{k^*} - \mu_{\hat{k}})$$

$$\begin{aligned} \mu_{k^*} &\in \left[\hat{\mu}_{k^*} - \sqrt{\frac{\ln(K/\delta)}{N}} \right] \\ \hat{\mu}_{k^*} &\in \left[\mu_{k^*} + \sqrt{\frac{\ln(K/\delta)}{N}} \right] \\ \mu_{\hat{k}} &\in \left[\mu_{\hat{k}} - \sqrt{\frac{\ln(K/\delta)}{N}} \right] \\ \mu_{k^*} - \mu_{\hat{k}} &\leq \left[\hat{\mu}_{k^*} + \sqrt{\frac{\ln(K/\delta)}{N}} \right] - \left[\hat{\mu}_{\hat{k}} - \sqrt{\frac{\ln(K/\delta)}{N}} \right] \end{aligned}$$



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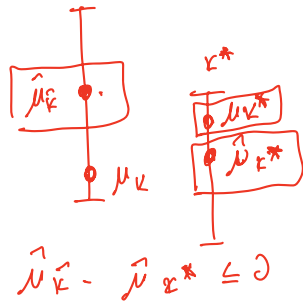
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$$\hat{\mu}_k \leq \hat{\mu}_{k^*}$$

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$$\leq 2\sqrt{\ln(K/\delta)/N}$$

$\hat{\mu}_{k^\star} \approx T - \text{length}$

$$\text{Regret}_{\text{exploit}} \leq (T - \underline{NK})(\mu_{k^\star} - \mu_{\hat{k}}) \leq \underline{2T} \sqrt{\frac{\ln(K/\delta)}{N}}$$

Finally, combine two regret together:

$$\text{Regret}_{\text{explore}} \leq N(K - 1) \leq NK$$

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$$\underbrace{\text{Regret}_T}_{\triangle} = \underbrace{\text{Regret}_{\text{explore}}}_{\triangle} + \underbrace{\text{Regret}_{\text{exploit}}}_{\triangle} \leq NK + 2T\sqrt{\frac{\ln(K/\delta)}{N}}$$

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Minimize the upper bound via optimizing N:

$$\text{Set } N = \underbrace{\left(\frac{T\sqrt{\ln(K/\delta)}}{K} \right)^{2/3}}, \text{ we have:}$$

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$$\text{Regret}_T \leq O\left(T^{2/3} K^{1/3} \cdot \ln^{1/3}(K/\delta)\right)$$

To conclude:

[Theorem] Fix $\delta \in (0,1)$, set $N = \left(\frac{T \sqrt{\ln(K/\delta)}}{2K} \right)^{2/3}$, with *# of time we pull each arm*, with probability at least $1 - \delta$, **Explore-Then-Commit** has the following regret:

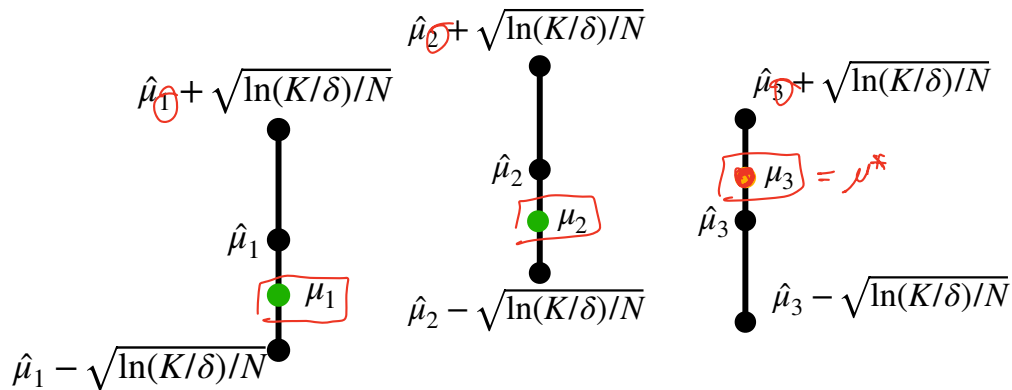
$$\text{Regret}_T \leq O \left(T^{2/3} K^{1/3} \cdot \ln^{1/3}(K/\delta) \right)$$

best = $T^{2/3}$, $K^{1/3} \rightarrow 0$ as $T \rightarrow \infty$

(See the MAB reading material for more details)

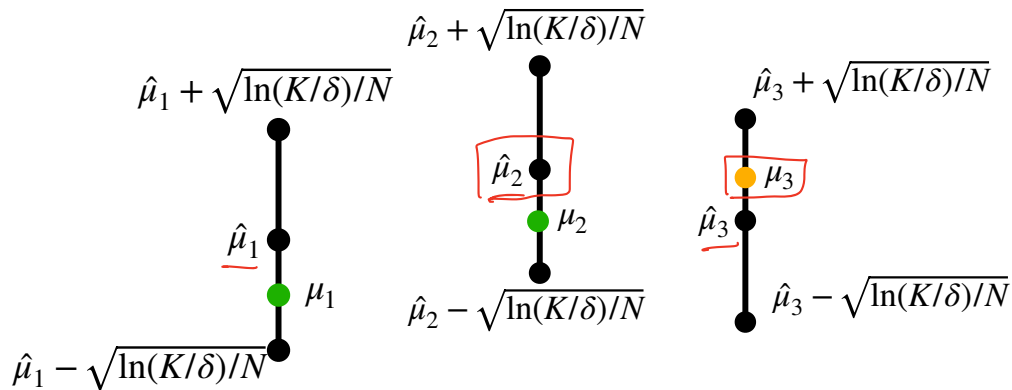
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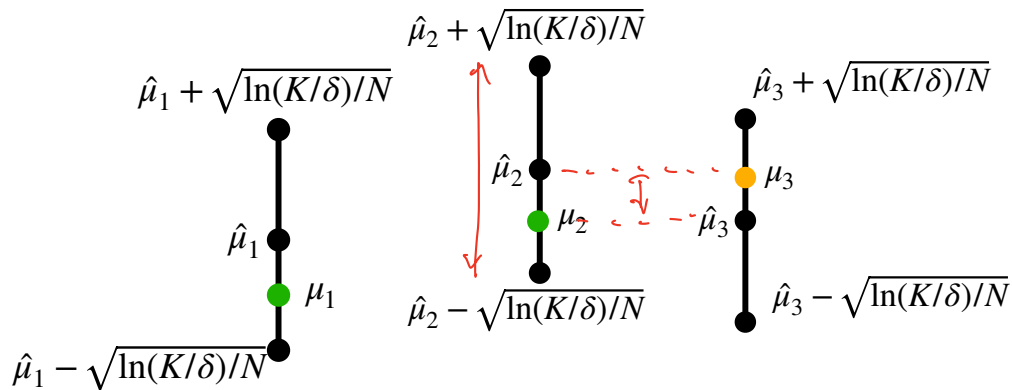
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(2): in the example above, we will commit to arm 2, and pay per-iter regret $(\mu_3 - \mu_2)$

But from the picture, we see that $(\mu_3 - \mu_2) \leq \text{length-of-Confidence-Interval}$

Question for Today: ETC has $T^{2/3}$

Can we design an algorithm that achieves $\widetilde{O}(\sqrt{T})$ regret?

Outline:

1. The upper Confidence Bound Algorithm
2. Analysis of UCB algorithm

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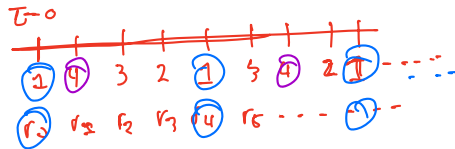
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$$\text{i.e., } \hat{\mu}_t(i) = \sum_{\tau=0}^{t-1} \mathbf{1}\{a_\tau = i\} \widehat{r_\tau} / N_t(i)$$

Assume we 4 arms



Recall the Tool for Building Confidence Interval:

[Hoeffding] Given a distribution $\nu \in \Delta([0,1])$, and N i.i.d samples $\{r_i\}_{i=1}^N \sim \nu$, w/ probability at least $1 - \delta$, we have:

$$\left| \underbrace{\sum_{i=1}^N r_i / N}_{\substack{\text{estimated} \\ \text{mean} \\ \hat{\mu}}} - \underbrace{\mu}_{\substack{\downarrow \\ \text{Ground} \\ \text{Truth}}} \right| \leq O\left(\sqrt{\frac{\ln(1/\delta)}{N}}\right)$$

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Thus, we know that for all iteration t , we have the for all $i \in [K]$, w/ prob $1 - \delta$,

$$|\hat{\mu}_t(i) - \mu_i| \leq \sqrt{\frac{\ln(1/(\delta/KT))}{N_t(i)}} \quad S = \frac{S}{KT}$$

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(Note that I applied union bound over all $t \in [T]$ and all $i \in [K]$, but let's not worry too much about log terms—*details are in reading material in case you are interested*)

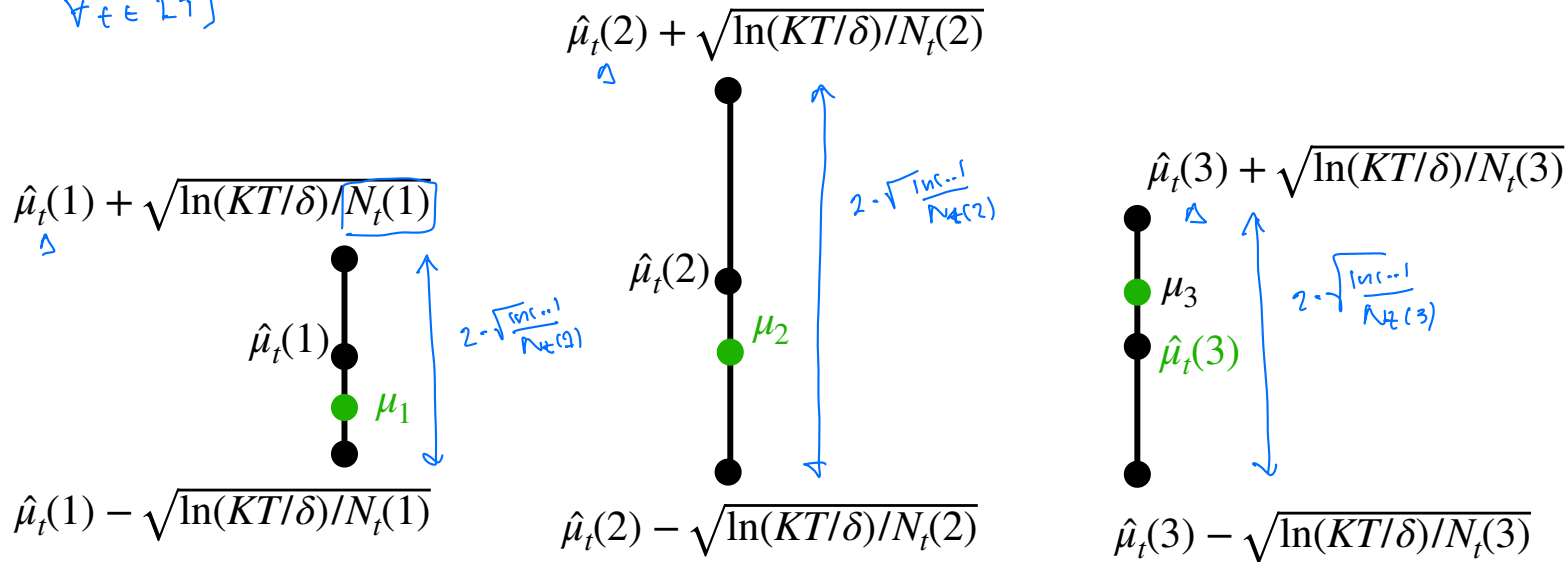
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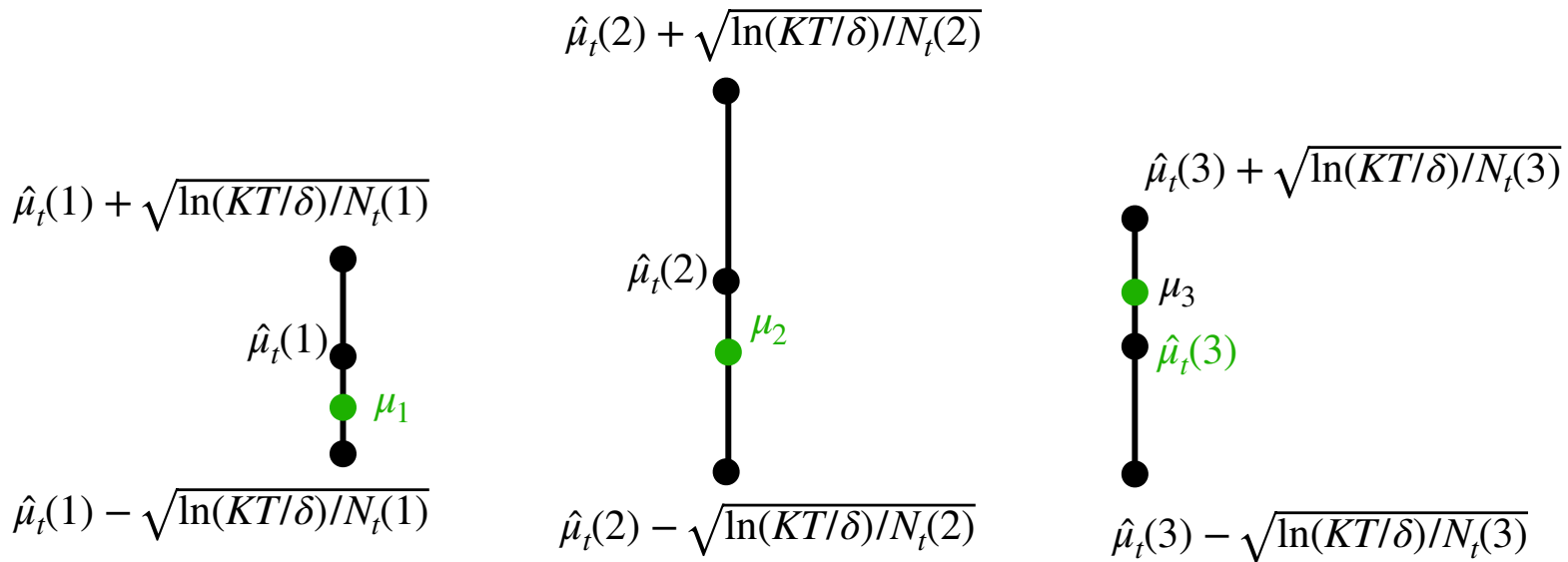
UCB: Optimism in the face of Uncertainty

Given the confidence interval, we pick arm that has the highest Upper-Conf-Bound:



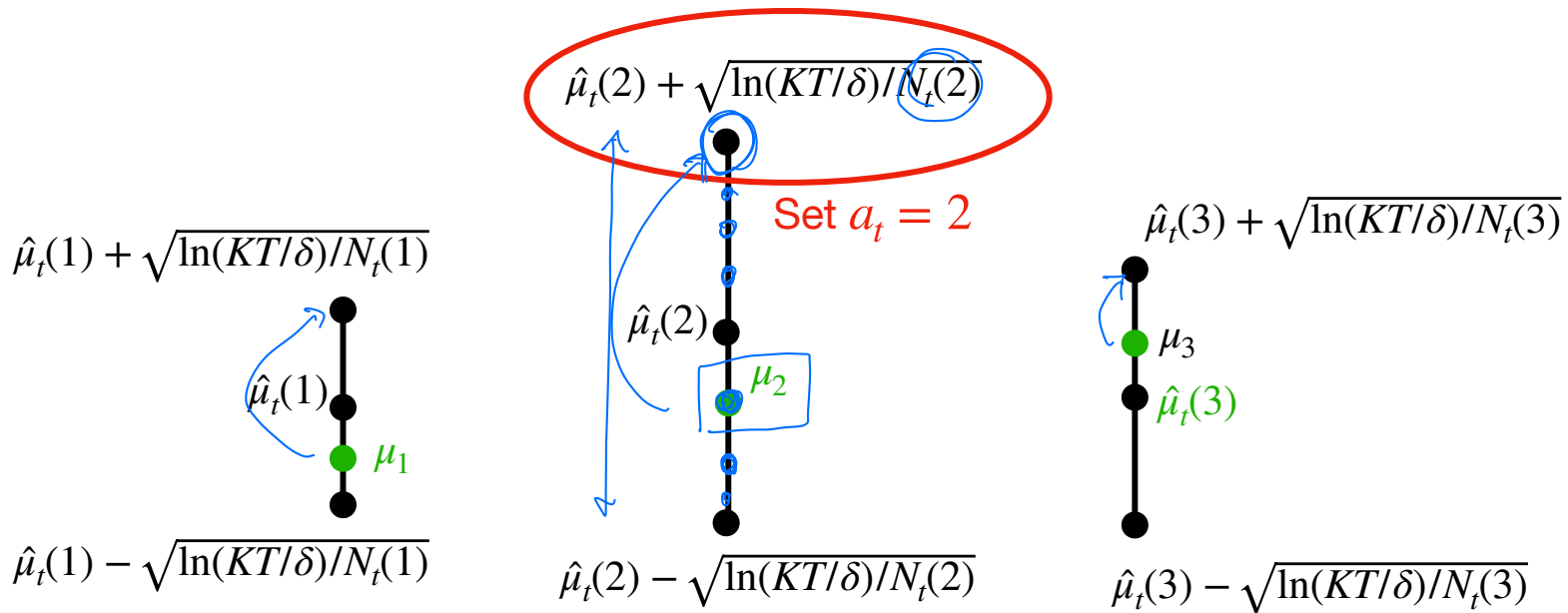
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Put things together: UCB Algorithm:

For $t = 0 \rightarrow T - 1$:

$$a_t = \arg \max_{i \in [K]} \left(\underbrace{\hat{\mu}_t(i)}_{\substack{\uparrow \\ \text{estimated} \\ \text{mean}}} + \underbrace{\sqrt{\frac{\ln(KT/\delta)}{N_t(i)}}}_{\substack{\uparrow \\ \text{half of the CI}}} \right)$$

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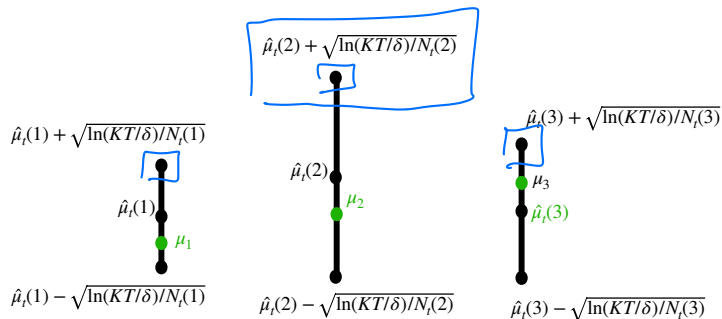
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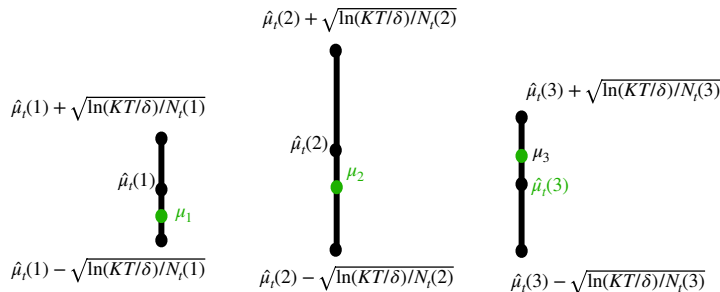


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“Reward Bonus”: $\sqrt{\frac{\ln(KT/\delta)}{N_t(i)}}$

Bonus $\uparrow$ if $N_t(i) \downarrow$
 Bonus $\downarrow$ if $N_t(i) \uparrow$

Outline:

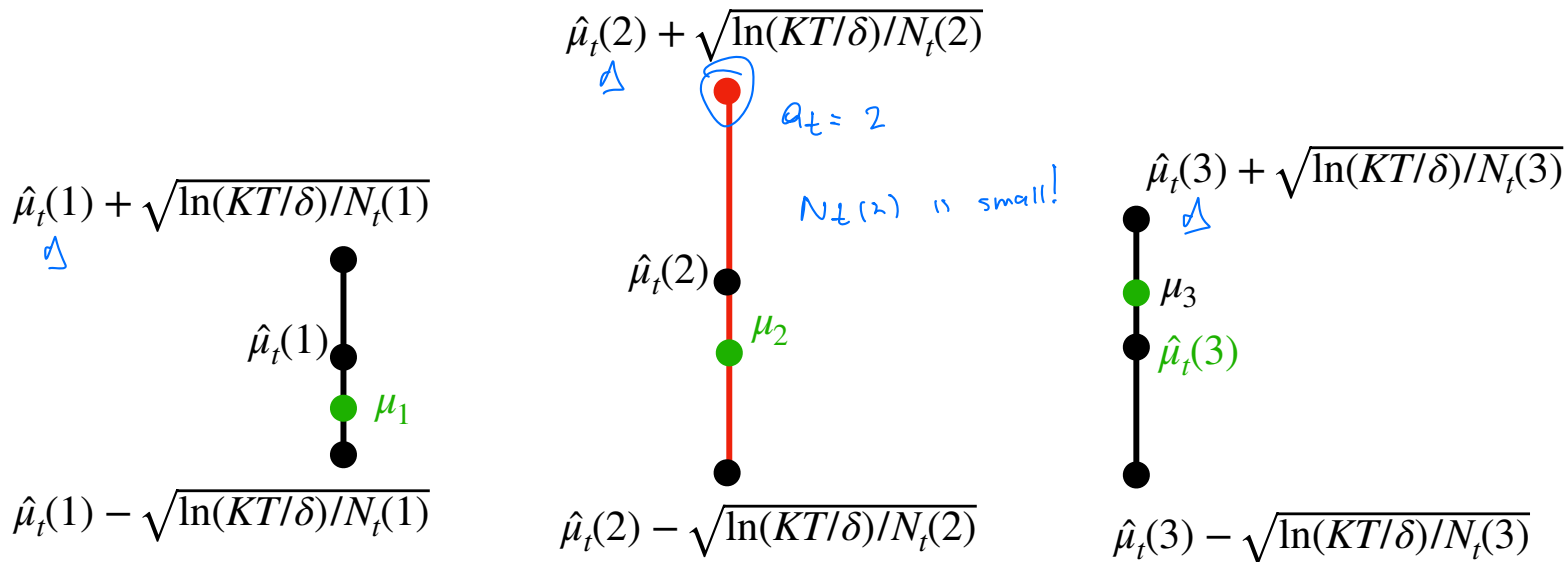


1. The upper Confidence Bound Algorithm

2. Analysis of UCB algorithm

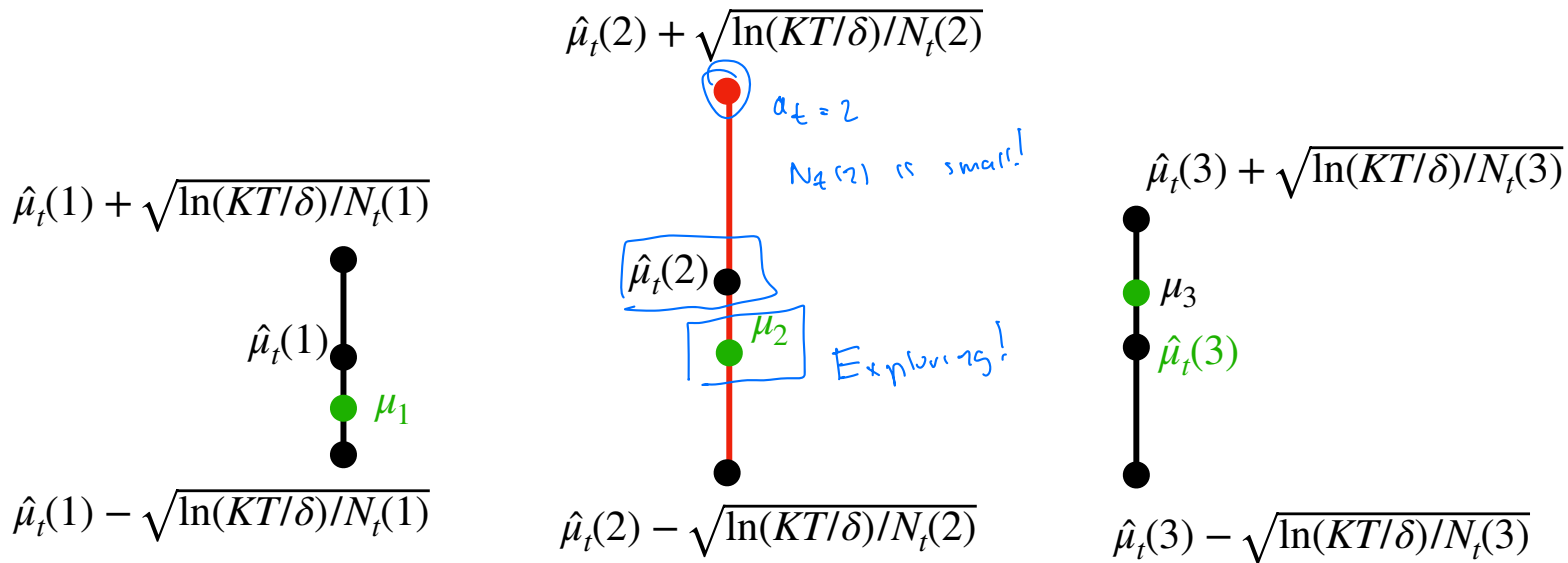
Intuitive Explanation of UCB

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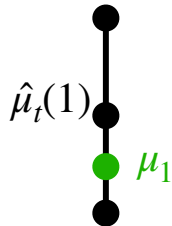
Case 1: it has large conf-interval, which means that it has not been tried many times yet (high uncertainty)



Intuitive Explanation of UCB

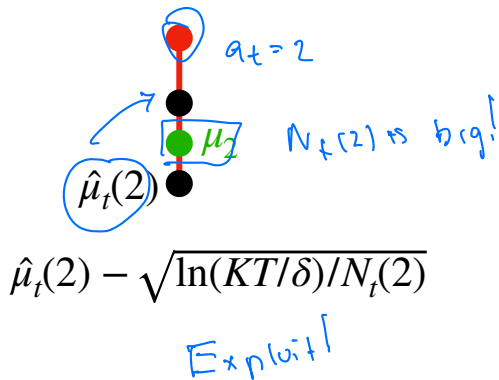
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$$\hat{\mu}_t(1) + \sqrt{\ln(KT/\delta)/N_t(1)}$$



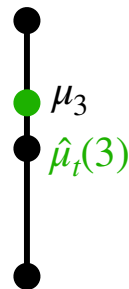
$$\hat{\mu}_t(1) - \sqrt{\ln(KT/\delta)/N_t(1)}$$

$$\hat{\mu}_t(2) + \sqrt{\ln(KT/\delta)/N_t(2)}$$



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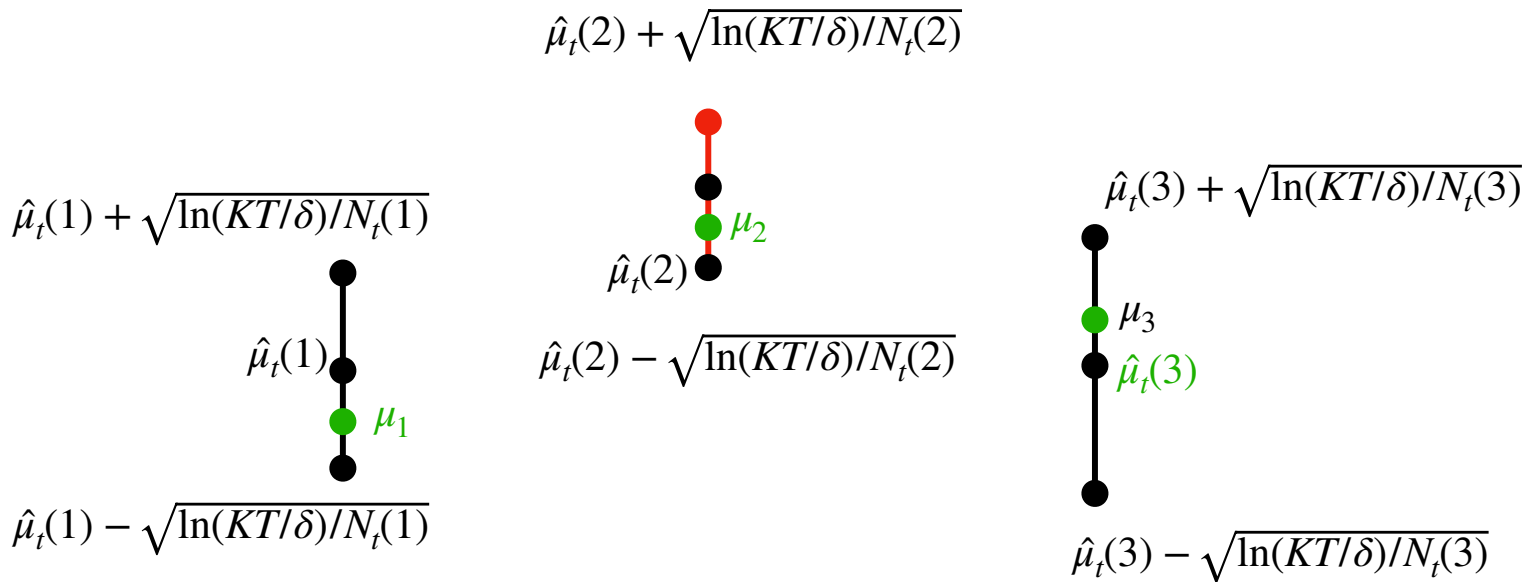
$$\hat{\mu}_t(3) + \sqrt{\ln(KT/\delta)/N_t(3)}$$



$$\hat{\mu}_t(3) - \sqrt{\ln(KT/\delta)/N_t(3)}$$

Intuitive Explanation of UCB

Case 2: it has low uncertainty, then it is simply a good arm, i.e., its true mean is high!



Exploration–Exploitation Tradeoff

Case 1: a_t has large conf-interval, which means that it has not been tried many times yet (high uncertainty)

Thus, we do exploration in this case!

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Case 2: a_t has small conf-interval, then it is simply a good arm, i.e., it's true mean is pretty high!

Thus, we do exploitation in this case!

△

Let's formalize the intuition

Denote the optimal arm $k^\star = \arg \max_{i \in [K]} \mu_i$; recall $a_t = \arg \max_{i \in [K]} \hat{\mu}_t(i) + \sqrt{\frac{\ln(KT/\delta)}{N_t(i)}}$

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$$\begin{aligned} \text{Regret-at-t} &= \mu^\star - \mu_{a_t} \\ &\leq \hat{\mu}_t(a_t) + \sqrt{\frac{\ln(TK/\delta)}{N_t(a_t)}} - \mu_{a_t} \end{aligned}$$

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$$\leq 2\sqrt{\frac{\ln(TK/\delta)}{N_t(a_t)}}$$

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Case 1: $N_t(a_t)$ is small

(i.e., uncertainty about a_t is large);

We pay regret, BUT we **explore** here,
as we just tried a_t at iter t !

Let's formalize the intuition

Denote the optimal arm $k^\star = \arg \max_{i \in [K]} \mu_i$; recall $a_t = \arg \max_{i \in [K]} \hat{\mu}_t(i) + \sqrt{\frac{\ln(KT/\delta)}{N_t(i)}}$

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Case 2: $N_t(a_t)$ is large, i.e., conf-interval of a_t is small,

Then we **exploit** here, as a_t is pretty good (the gap between $\mu^\star$ & μ_{a_t} is small)!

Let's formalize the intuition

Finally, let's add all per-iter regret together:

$$\begin{aligned}\text{Regret}_T &= \sum_{t=0}^{T-1} \left(\mu^\star - \mu_{a_t} \right) \\ &\leq \sum_{t=0}^{T-1} 2\sqrt{\frac{\ln(TK/\delta)}{N_t(a_t)}} \\ &= 2\sqrt{\ln(TK/\delta)} \cdot \sum_{t=0}^{T-1} \sqrt{\frac{1}{N_t(a_t)}}\end{aligned}$$

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Lemma (optional):

$$\sum_{t=0}^{T-1} \sqrt{\frac{1}{N_t(a_t)}} \leq O\left(\sqrt{KT}\right)$$

UCB Regret:

[Theorem (informal)] With high probability, UCB has the following regret:

$$\text{Regret}_T = \widetilde{O}\left(\sqrt{KT}\right)$$

(See reading material for more details)

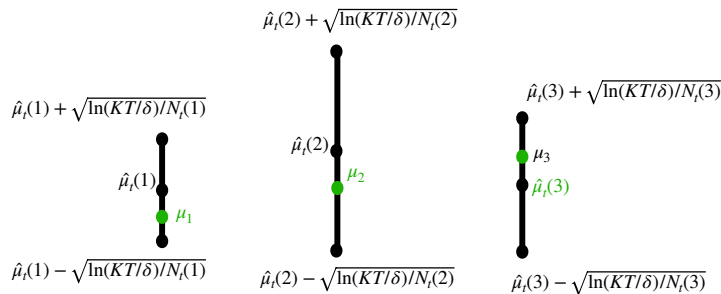
Summary for Today:

UCB algorithm: *Principle of Optimism in the face of Uncertainty*

For $t = 0 \rightarrow T - 1$:

$$a_t = \arg \max_{i \in [K]} \left(\hat{\mu}_t(i) + \sqrt{\frac{\ln(KT/\delta)}{N_t(i)}} \right)$$

(# Upper-conf-bound of arm i)



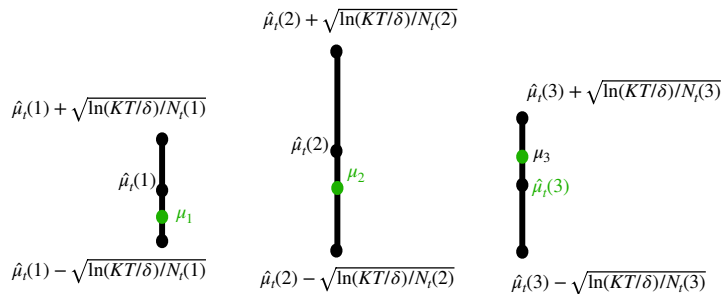
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Analysis Intuition:

Case 1: the arm a_t has high uncertainty (we explore)

Case 2: the arm a_t has low uncertainty, then it must be a near-optimal arm (i.e., exploit)

Feedback and Attendance

<https://bit.ly/4gYn2nC>

